

Plan: Extend cq-level Range from [0, 63] to [0, 80]

Context

In AOM_Q (constant quality) mode, the user-facing cq-level parameter (0-63) maps to internal qindex values (0-255) via a lookup table. Due to quality boost multipliers applied per hierarchical layer (KF/ARF via qstep_ratio, intermediate ARF via midpoint interpolation), cq-level=63 only produces qindex ~173 for key frames. The full qindex range [0, 255] is unreachable for boosted frame types.

Extending cq-level to 80 allows all frame types -- including key frames -- to reach qindex 255, giving the encoder full quantizer coverage across all hierarchical layers.

This feature is only needed in constant quality mode (end-usage=q). Bitrate control modes (CBR, VBR) are unaffected.

Scope

The extension must work regardless of the use-fixed-qp-offsets setting:

- use-fixed-qp-offsets=1: Uses qstep_ratio approach in encoder_utils.c -- covered in Part 1
- use-fixed-qp-offsets=0: Uses polynomial minq tables / av1_compute_qdelta in ratectrl.c -- covered in Part 2

The behavior differs between the two paths because they use different formulas to derive per-layer qindex from the base cq-level.

Part 1: Fixed QP Offsets Path (use-fixed-qp-offsets=1)

Approach: Virtual qindex > 255 with Linear qstep_ratio Ramp

Store unclamped "virtual" qindex values (259-323) in oxcf->cq_level for cq-levels 64-80.

For the use_fixed_qp_offsets=1 code path in encoder_utils.c, a unified qstep_ratio approach is used across the entire cq-level range:

- cq 0-63 (qindex 0-255): qstep_ratio = $0.2 + (1.0 - \text{awq}/255) * \text{qratio_grad}$ -- ratio decreases from 0.5 to 0.2 as quality decreases (existing formula, unchanged)
- cq 64-80 (virtual qindex 259-323): qstep_ratio = $0.2 + (\text{cq_level} - 255) / (323 - 255) * 0.8$ -- ratio increases from 0.2 toward 1.0, so KF/ARF qindex converges to MAXQ

At cq=63, the ratio reaches 0.2 (minimum boost). For cq 64-80, the ratio smoothly ramps back toward 1.0, meaning KF/ARF qindex approaches the leaf qindex (255). At cq=80, all layers converge to qindex 255.

When qstep_ratio >= 1.0 (which occurs at cq=80), the av1_get_q_index_from_qstep_ratio() search is bypassed and min(cq_level, MAXQ) is returned directly, ensuring exact convergence to qindex 255 without off-by-1 rounding from the DC quantizer lookup.

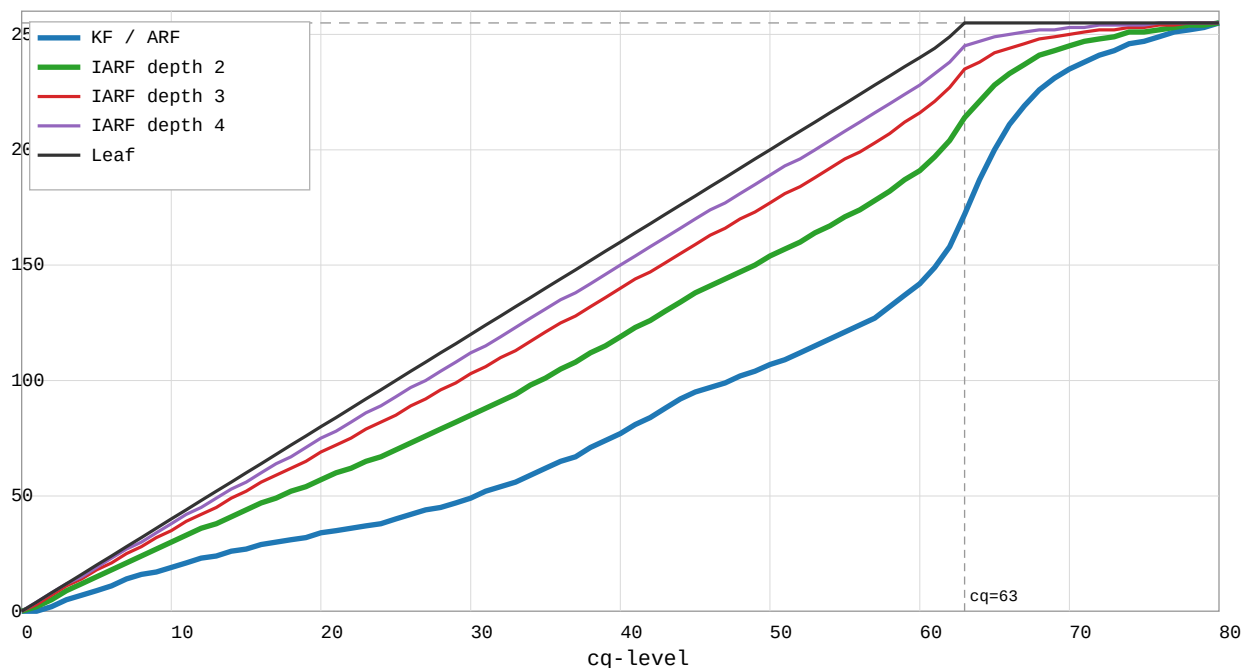
Verified qindex mapping (10-bit, use-fixed-qp-offsets=1)

Values verified against actual encoder output. At cq <= 79, KF/ARF shows off-by-1 from predicted values due to DC quantizer rounding; IARF and Leaf are exact. At

cq=80, the qstep_ratio >= 1.0 shortcut ensures all layers reach exactly 255.

cq	qindex	KF/ARF	IARF_d2	IARF_d3	IARF_d4	Leaf
55	220	122	171	196	208	220
60	240	142	191	216	228	240
63	255	172	214	235	245	255
64	259	187	221	238	247	255
66	267	212	234	245	250	255
68	275	227	241	248	252	255
70	283	235	245	250	253	255
72	291	242	249	252	254	255
74	299	247	251	253	254	255
76	307	250	253	254	255	255
78	315	253	254	255	255	255
79	319	254	255	255	255	255
80	323	255	255	255	255	255

cq-level to qindex (use-fixed-qp-offsets=1)



Changes for Fixed QP Path

1. Extend quantizer_to_qindex table and reverse function

File: av1/encoder/av1_quantize.c

- Update comment: "0-63" to "0-80"
- Add 17 entries continuing from 255 with step +4 (unclamped virtual values):
259, 263, 267, 271, 275, 279, 283, 287, 291,
295, 299, 303, 307, 311, 315, 319, 323,
- av1_qindex_to_quantizer(): change loop bound < 64 to < 81, fallback return 63 to return 80
- Total table size: 81 entries (cq 0-80)

2. Clamp best_allowed_q and worst_allowed_q

File: av1/av1_cx_iface.c

best_allowed_q and worst_allowed_q feed directly into quantization and must stay in [0, 255]. cq_level is only used in rate control delta math and can exceed 255:

```
oxcf->best_allowed_q =
    extra_cfg->lossless ? 0 : AOMMIN(av1_quantizer_to_qindex(cfg->rc_min_quantizer), MAXQ);
oxcf->worst_allowed_q =
    extra_cfg->lossless ? 0 : AOMMIN(av1_quantizer_to_qindex(cfg->rc_max_quantizer), MAXQ);
oxcf->cq_level = av1_quantizer_to_qindex(extra_cfg->cq_level); // can be > 255
```

3. Guard av1_convert_qindex_to_q() against qindex > 255

File: av1/encoder/ratectrl.c

Add clamp at the top of the function so ac_qlookup[] is not accessed out of bounds:

```
double av1_convert_qindex_to_q(int qindex, aom_bit_depth_t bit_depth) {
    qindex = AOMMIN(qindex, MAXQ); // clamp to valid ac_qlookup range
    // ... rest unchanged
}
```

4. Update range checks

File: av1/av1_cx_iface.c

Location | Current | New | Notes

-----+-----+-----+-----

RANGE_CHECK_HI(extra_cfg, cq_level, 63) | 63 | 80 | Allow extended cq-level

Not changed (decision deferred):

- RANGE_CHECK_HI(cfg, rc_max_quantizer, 63) -- stays at 63 (used by CBR/VBR)
- 63, // rc_max_quantizer default -- stays at 63

5. Update fixed QP offsets formula for extended range

File: av1/encoder/encoder_utils.c

Update max_virtual_qindex from 327 to 323, and add a qstep_ratio >= 1.0 shortcut to bypass av1_get_q_index_from_qstep_ratio() when the ratio means no boost:

```
if (cq_level <= MAXQ) {
    qstep_ratio = 0.2 + (1.0 - (double)cpi->rc.active_worst_quality / MAXQ) * qratio_grad;
} else {
    const int max_virtual_qindex = 323; // quantizer_to_qindex[80]
    qstep_ratio = 0.2 + (double)(cq_level - MAXQ) / (max_virtual_qindex - MAXQ) * 0.8;
}
if (qstep_ratio >= 1.0)
    *q = AOMMIN(cq_level, MAXQ);
else
    *q = av1_get_q_index_from_qstep_ratio(
        cpi->rc.active_worst_quality, qstep_ratio,
        cm->seq_params->bit_depth);
```

Why the shortcut is needed: av1_get_q_index_from_qstep_ratio() searches downward from leaf_qindex for a qindex whose DC qstep satisfies qstep + 0.1 <= target_qstep. When qstep_ratio = 1.0, target_qstep = leaf_qstep, so leaf_qstep + 0.1 <= leaf_qstep is false and the search drops to leaf_qindex - 1 (254 instead of 255). The shortcut avoids this off-by-1, ensuring all layers converge to exactly 255 at cq=80.

The shortcut only activates at cq=80 where the ratio reaches exactly 1.0. For all other cq levels (0-79), the ratio is < 1.0 and the normal search path is used.

6. No changes needed

- aomenc.c: CLI option definitions have no hardcoded range -- validation is in

av1_cx_iface.c

- warnings.c: check_quantizer() only warns about narrow Q range
- QINDEX_RANGE, MAXQ, ac_qlookup[]: Internal qindex space [0, 255] is unchanged
- Rate control minq tables: Indexed by qindex [0, 255], unchanged
- Existing clamp() calls in rc_pick_q_and_bounds_*: Already clamp final q to [rc->best_quality, rc->worst_quality] = [0, 255]

Part 2: Non-Fixed QP Path (use-fixed-qp-offsets=0)

How it works

In AOM_Q mode without fixed QP offsets, per-layer qindex is derived through these code paths:

Frame Type | Dispatch | Formula

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KF (first) | rc_pick_q_and_bounds_no_stats | $cq_level + av1_compute_qdelta(q_val, q_val * 0.25)$

KF (subseq.) | get_intra_q_and_bounds | get_kf_active_quality(active_worst_quality)

ARF/GF | get_active_best_quality | $cq_level + av1_compute_qdelta(q_val, q_val * 0.40)$

IARF | get_active_best_quality | Midpoint interpolation from arf_q toward cq_level

Leaf/Overlay | get_active_best_quality | return cq_level directly

Natural extension behavior

For extended $cq > 63$, the `av1_compute_qdelta` approach produces a naturally smooth linear ramp without any special formula:

- $q_val = av1_convert_qindex_to_q(cq_level)$ clamps to $MAXQ=255$, so q_val is constant (457.0 for 10-bit)
- The delta is therefore constant: -47 for ARF (factor 0.40)
- $active_best = clamp(cq_level + delta, 0, 255)$ increases by +4 per cq step until convergence
- Final q is clamped to [0, 255]

This means each +1 on cq translates to +4 on actual qindex until the clamp kicks in.

Verified qindex mapping (10-bit, use-fixed-qp-offsets=0, actual encoder)

Values from actual encoder output after the first-ARF bug fix. ARF qindex is now consistent across all GOPs. KF varies slightly between first KF and subsequent KFs due to state adaptation.

cq | qindex | KF (range) | ARF | IARF_d2 | IARF_d3 | IARF_d4 | Leaf

---+-----+-----+-----+-----+-----+-----

55 | 220 | ~135 | 173 | 197 | 209 | 215 | 220

60 | 240 | 155-159 | 192 | 216 | 228 | 234 | 240

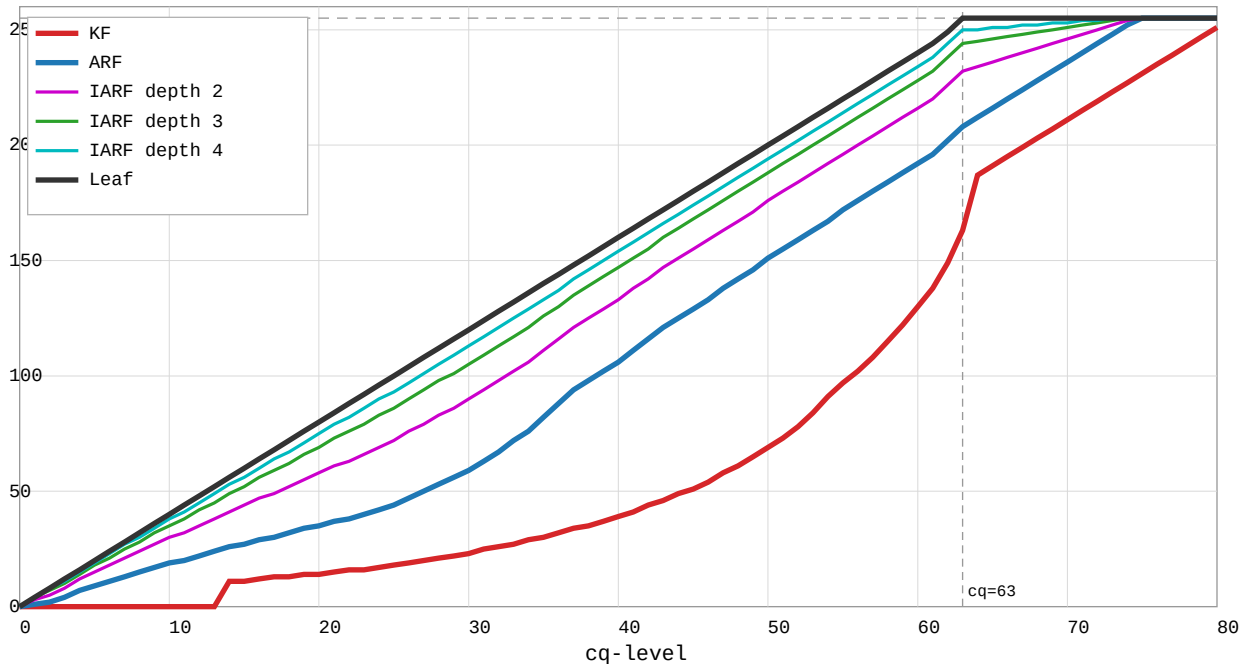
63 | 255 | 171-185 | 208 | 232 | 244 | 250 | 255

64 | 259 | 175-187 | 212 | 236 | 248 | 254 | 255

70 | 283 | 200-211 | 236 | 255 | 255 | 255 | 255

80 | 323 | 219-251 | 255 | 255 | 255 | 255 | 255

cq-level to qindex (use-fixed-qp-offsets=0)



Convergence at max_cq=80

With max_cq=80 (virtual qindex=323):

- ARF: reaches 255 (fully converged)
- IARF/Leaf: converged at 255
- KF: reaches ~219-251 depending on content/state (first KF vs subsequent KFs)

KF does not fully reach 255 at cq=80 in the non-fixed QP path, but gets close enough (~219-251). This is acceptable since the non-fixed QP path is inherently content-adaptive, and KF quality is state-dependent by design.

First-ARF Bug: Root Cause and Fix

Bug: When `cq_level > 255` and `use-fixed-qp-offsets=0`, the first ARF frame in a sequence got `qindex=0`, causing massive bitrate inflation (e.g., 1,077 kbps at `cq=64` vs expected ~69 kbps).

Root cause: In AOM_Q mode, non-ARF_UPDATE frames are dispatched via `rc_pick_q_and_bounds()` -> `rc_pick_q_and_bounds_q_mode()` -> `get_active_best_quality()`. For ARF/GF frames (non-leaf, non-IARF), the code used `get_gf_active_quality()` -- a boost-factor-based minq lookup designed for VBR/CQ modes. This function ignores `cq_level` entirely and computes quality based on `avg_frame_qindex`, which is very low early in the sequence (dominated by the high-quality KF). This produced near-zero qindex for the first ARF, which then cascaded into bad IARF values via midpoint interpolation.

The second and subsequent ARFs appeared to work because by that point `avg_frame_qindex[INTER_FRAME]` had been updated to reasonable values, masking the underlying issue.

Fix: In `get_active_best_quality()` (`ratectrl.c`), add an early return for `rc_mode == AOM_Q` non-IARF frames that uses the `cq_level + delta` formula:

```
if (rc_mode == AOM_Q && !is_intrl_arf_boost) {
    const double q_val = av1_convert_qindex_to_q(cq_level, bit_depth);
```

```

const int delta_qindex =
    av1_compute_qdelta(rc, q_val, q_val * 0.40, bit_depth);
active_best_quality =
    clamp(cq_level + delta_qindex, rc->best_quality, rc->worst_quality);
return active_best_quality;
}

```

This ensures: ARF/GF qindex is derived from cq_level (consistent with AOM_Q design); the delta is constant for cq > 63, producing a smooth linear ramp; clamping to [0, 255] provides natural convergence; IARF frames still use p_rc->arf_q with midpoint interpolation (unchanged).

Bitrate validation (non-fixed QP, after fix)

Bitrate now decreases monotonically with increasing cq-level:

cq	Bitrate (bps)	PSNR-Y (dB)	Before fix (bps)
60	100,246	32.16	137,435
63	72,612	30.19	99,891
64	68,676	29.87	1,077,227
70	47,294	27.81	493,705
80	31,805	25.19	307,166

Safety Analysis: Impact on Other Rate Control Modes

Mode	Uses cq_level?	Impact	Status
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AOM_CBR	No	Completely unaffected	SAFE
AOM_VBR	No	Completely unaffected	SAFE
AOM_CQ	Yes (comparisons)	cq_level in if (q < cq_level) -- final q clamped [0,255]	SAFE
AOM_Q (fixed QP)	Yes (direct)	Extended range handled by qstep_ratio ramp + shortcut	SAFE with fixes
AOM_Q (non-fixed)	Yes (direct)	ARF uses cq_level + delta formula; linear ramp	SAFE with fix

Key safety guarantees:

- av1_ac_quant() in quant_common.c already has internal clamp(qindex + delta, 0, MAXQ) -- defense-in-depth
- All rc_pick_q_and_bounds_* functions clamp final q to [rc->best_quality, rc->worst_quality] = [0, 255]
- best_allowed_q/worst_allowed_q flow into rc->worst_quality used as array index for inter_minq[]/rtc_minq[] (256 elements) -- must stay <= 255 (Fix #2)
- cq_level is never used as a direct array index -- only in arithmetic and as input to av1_convert_qindex_to_q() (Fix #3)
- The qstep_ratio >= 1.0 shortcut in encoder_utils.c only activates at cq=80 and returns AOMMIN(cq_level, MAXQ) = 255
- The AOM_Q early return in get_active_best_quality() clamps to [rc->best_quality, rc->worst_quality] = [0, 255]

Files Modified

File	Changes	Status
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av1/encoder/av1_quantize.c	Extend table to 81 entries; update reverse lookup	Done
av1/av1_cx_iface.c	cq_level range 63->80; clamp best/worst_allowed_q	Done
av1/encoder/ratectrl.c	Clamp in av1_convert_qindex_to_q(); fix ARF in get_active_best_quality()	Done
av1/encoder/encoder_utils.c	max_virtual_qindex 327->323; qstep_ratio>=1.0 shortcut	Done

Test Configuration

The primary test configuration uses `use_fixed_qp_offsets=1` with `deltaq-mode=0` to achieve deterministic, content-independent qindex assignment per hierarchical layer.

Encoding Command

```
aomenc --verbose --codec=av1 -v --psnr --obu \
--frame-parallel=0 --cpu-used=0 --limit=600 --passes=1 --end-usage=q \
--i420 --use-fixed-qp-offsets=1 --deltaq-mode=0 --enable-tpl-model=0 \
--fps=60/1 --input-bit-depth=10 --bit-depth=10 --cq-level=$CQ \
--tile-columns=0 --threads=1 --enable-keyframe-filtering=1 \
--kf-min-dist=65 --kf-max-dist=65 --min-gf-interval=16 --max-gf-interval=16 \
--gf-min-pyr-height=4 --gf-max-pyr-height=4 --lag-in-frames=19 --auto-alt-ref=1 \
-w 1920 -h 1080 \
-o output.obu input.y4m
```

Key flags explained

Flag | Value | Purpose

```
-----+-----+-----
--use-fixed-qp-offsets | 1 | Bypass polynomial minq lookup; use qstep_ratio
--deltaq-mode | 0 | Disable per-SB delta-q adaptation
--enable-tpl-model | 0 | Disable TPL; prevent override of fixed qp offsets
--passes | 1 | Single-pass; uses rc_pick_q_and_bounds_no_stats for ARF
--kf-min/max-dist | 65 | Fixed keyframe interval at every 65 frames
--min/max-gf-interval | 16 | Fixed golden frame interval at 16 frames
--gf-min/max-pyr-height | 4 | 4-level pyramid (ARF->IARF_d2->d3->d4->Leaf)
--bit-depth | 10 | 10-bit encoding; uses dc_qlookup_10_QTX[]
```

Q selection code path with this configuration

With `use_fixed_qp_offsets=1`, the final qindex override happens in `av1_set_size_dependent_vars()` (`encoder_utils.c`), which runs after the initial `av1_rc_pick_q_and_bounds()` selection:

- KF/ARF/GF frames (`is_frame_tpl_eligible()` returns true):
 - `cq_level <= 255`: `qstep_ratio = 0.2 + (1.0 - awq/255) * qratio_grad`, search downward via `av1_get_q_index_from_qstep_ratio()`
 - `cq_level > 255` (`cq` 64-80): `qstep_ratio = 0.2 + (cq_level - 255) / (323 - 255) * 0.8`, same function with `leaf_qindex = 255`
 - When `qstep_ratio >= 1.0` (`cq=80`): shortcut returns `min(cq_level, 255) = 255` directly
 - KF and ARF use the same `qstep_ratio` (unified, no separate boost factors)
- INTNL_ARF frames: midpoint interpolation $q = (\text{arf_q} + \min(\text{cq_level}, 255) + 1) / 2$ per depth level
- Leaf/Overlay frames: $q = \min(\text{cq_level}, 255)$

This produces a deterministic qindex-per-layer mapping that depends only on `cq_level`.

Verification

Fixed QP path (`use-fixed-qp-offsets=1`)

Test | Expected Result | Status

```
-----+-----+-----
cq-level=32 | No regression | Pass
cq-level=70 | Valid output, per-layer qindex differentiation | Pass
cq-level=80 | All layers reach qindex 255 | Pass
```

cq-level=81 | Rejected with error | Pass

Bitrate (480x270, 130 frames, cpu-used=4):

cq | Bitrate (bps) | PSNR-Y (dB)

-----+-----+-----

60 | 154,870 | 34.38

63 | 100,685 | 31.58

64 | 82,633 | 30.28

70 | 39,651 | 25.98

80 | 27,969 | 23.63

Non-fixed QP path (use-fixed-qp-offsets=0)

Test | Expected Result | Status

-----+-----+-----

cq-level=32 | No regression | Pass

cq-level=64 | No qindex=0 anomaly (first-ARF fix) | Pass

cq-level=70 | Valid output, monotonic bitrate decrease | Pass

cq-level=80 | ARF/IARF/Leaf reach qindex 255 | Pass

Bitrate (480x270, 130 frames, cpu-used=4):

cq | Bitrate (bps) | PSNR-Y (dB)

-----+-----+-----

60 | 100,246 | 32.16

63 | 72,612 | 30.19

64 | 68,676 | 29.87

70 | 47,294 | 27.81

80 | 31,805 | 25.19

Other modes

Test | Expected Result | Status

-----+-----+-----

CBR mode (--end-usage=cbr) | No regression (cq_level not used) | Pass

CQ mode (--end-usage=cq --cq-level=70) | Runs correctly | Pass

Unit tests (make runtests) | All pass | --

Reference

- Extended range plot (fixed QP): cq_extended_fixed_maxcq80.svg
- Unified plot: qindex_per_layer_chart_unified.svg
- Original range comparison: cq_to_qindex_fixed.svg / cq_to_qindex_nonfixed.svg
- Alternative max_cq options: cq_extended_fixed_maxcq70.svg, cq_extended_fixed_maxcq75.svg
- Data: cq_to_qindex_data.csv
- Computation script: plot_cq_to_qindex.py